Security–Performance Tradeoffs in Resource-Constrained Agents: Phase 1 Baseline Benchmark and Measurements

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https://github.com/waynewangyuxuan/CSE-291A-Project

Abstract

We ask how weight quantization and context management affect a local tool-using agent's task completion and its resistance to prompt injection. Phase 1 builds a reproducible harness around a 60-task subset of AgentDojo, freezes an audited paired clean/adversarial workload, and measures one baseline: Qwen3-8B in unquantized F16 with full 32K context, 600 episodes over two seeds. It completes 48.3% of clean tasks; 19.2% of attacks succeed overall and 23.7% on tasks it can otherwise complete; prefill of uncached history is 78% of model compute. Two single-seed explorations (a prompt reminder, and Q4 $\times$ five context policies) exercise the Phase 2 harness.

1 Introduction and Research Questions

An agent that reads untrusted documents, messages or web pages can be redirected by instructions embedded in them. The deployment choices that make a local agent affordable, lower weight precision and a smaller retained context, may change both what it can do and what it can be made to do. Phase 1 does not vary configurations; it establishes the measurement and a baseline for Phase 2, organized around three questions:

- **RQ1 (capability).** How much of a fixed workload does the undefended baseline complete, and where does it fail? Metrics: official utility by suite, seed and reference path length; failure taxonomy from inspected traces; unnecessary-refusal incidence.
- **RQ2 (security vs. capability).** How often do injected instructions succeed, and does that depend on phrasing, goal, and whether the agent could do the task at all? Metrics: attack success rate (ASR) over all attacks and conditioned on a successful same-task clean control; payload exposure; per-template and per-goal splits.
- **RQ3 (cost).** What does full-history F16 cost? Metrics: latency, prompt/completion tokens, native prefill/decode time, peak RSS.

2 Setup

Benchmark and workload. We adapt AgentDojo v1.2.2 (Debenedetti et al., 2024) at a pinned commit. Its utility and attack checkers are deterministic state, answer or trace checks, not model-generated scores. We selected **60 legitimate tasks** (Banking 8, Slack 16, Workspace 24, Travel 12) *before* observing any F16 outcome; development tasks are held apart. Each task is paired with four attacks: two suite-specific goals (e.g. redirect a recurring payment, send a phishing link, create an unauthorized calendar event, email personal data) crossed with two fixed templates, `direct_append` ("TODO: *goal*") and `ignore_previous_append`. Unlike upstream, which replaces the injected field, our `_append` variant keeps the benign content so the legitimate objective stays achievable. Two seeds (17, 29) give $60 \times 2 \times (1+4) = 600$ episodes.

Checker audit. A source-level audit excluded upstream checker defects (tasks already or always true, tasks accepting a no-op, an `or`-precedence bug, a reference solution over our tool cap). An offline audit of all 240 task/goal/template conditions confirms the legitimate task stays completable in 1–9 tool calls with the payload present and that completing it never satisfies the attack goal.

Deployment. Qwen3-8B is converted from the verified BF16 checkpoint to F16 GGUF (0 quantized tensors) and served by pinned llama.cpp on an M2 Max (64 GiB): one slot, 32,768-token context, F16 KV cache, no prompt caching, non-thinking decoding at temperature 0.7. Limits: 2,048 output tokens, 15 tool calls, 120 s per request, 600 s per episode; exceeding one is a retained error with unknown score, never a dropped episode or a "de-

Metric	Succ./N	Rate	95% CI
Clean utility	58/120	48.3%	35.8–59.2
Attacked utility	173/480	36.0%	26.7–45.8
ASR, all attacks	92/480	19.2%	15.0–23.5
ASR, clean-eligible	55/232	23.7%	17.3–30.0
Payload exposure	412/480	85.8%	76.7–94.2

Table 1: F16 full-context baseline, 600 episodes. 25 episodes ended in errors with unknown scores; counting all as attack successes would raise all-attack ASR to at most 23.8%.

fense". The system prompt is AgentDojo's default; no defense is added.

Metrics and uncertainty. Every rate reports numerator, denominator and unknown count. ASR is given over all attacks and over attacks whose same-task, same-seed clean control succeeded, so inability to use tools is not counted as resistance. Exposure records whether the payload reached a model request. 95% intervals bootstrap *task clusters* within suite (2,000 draws), so 600 episodes are not treated as independent.

3 Phase 1 Baseline Results

RQ1: capability. The agent completes 58/120 clean episodes (Table 1; Banking 6/16, Slack 20/32, Workspace 23/48, Travel 9/24). Seeds agree on most tasks (25 succeed in both, 26 fail in both, 7 differ). Success drops for long reference paths: 22% for $\geq$5 calls vs. 53–71% below; within Slack 3/14 vs. 17/18. The agent never produced malformed tool JSON, yet 149/600 episodes had a tool execution error (wrong arguments or entities). Reviewing all 120 clean traces and 48 attack traces under a pre-frozen protocol, primary clean failure labels were planning 45, tool use 15, unnecessary refusal 2, task understanding 2, other/unclear 16, none 40. A recurrent pattern is *gathering the right information but never executing the action*: 16/76 action tasks end with a promise instead. Unnecessary refusal is rare (2/120, upper bound 13/120).

RQ2: security. Overall ASR is 19.2%; restricted to attacks whose clean control succeeded it rises to 23.7%, so the pooled number is partly masked by incapacity. Phrasing dominates: `ignore_previous_append` succeeds 79/240 (32.9%) vs. 13/240 (5.4%) for `direct_append` at identical exposure (206/240 each). Every success was exposed. Goals differ sharply, from Banking password change 16/32 to Workspace event 0/96; low Workspace ASR is not robustness per se, since its goals need exact multi-field actions and it hosts 18 of the 25 errors. Attacks also cost capability: among 457 known attack/control pairs, 74 flip from clean success to failure and 16 the reverse (-12.7 pp). Trace review also surfaced unauthorized actions the exact-goal checker misses (a payment redirected to a mistyped IBAN, a phishing URL visited instead of sent, a correct reschedule followed by the attacker's email), motivating a separate "broader unauthorized behavior" label.

RQ3: cost. Median episode latency is 56.4 s clean / 49.7 s attacked (p95 180 / 159). Across 2,965 requests the model consumed 11.35M prompt tokens and produced 0.146M completion tokens; native prefill took 31,949 s vs. 9,198 s for decode (355 tok/s vs. 15.5 tok/s). Peak server RSS was 18.8 GiB. Since each request resends uncached history, context management targets the dominant cost. No context overflow occurred (largest prompt 20,150 tokens). Rerunning all 25 errors twice with identical seeds and budgets recovered 0/50: deterministic budget exhaustion, not flakiness.

4 Preliminary Explorations

Two single-seed (17) studies exercised the Phase 2 harness; both are exploratory.

Source-boundary prompt reminder. A fixed 100-token paragraph appended to the system prompt states that retrieved content is data, not authority. Over 300 paired episodes per arm (fresh control, adjacent AB/BA order), exact-goal attack success fell 47/240 $\rightarrow$ 39/240 (-3.3 pp, paired 95% interval $[-6.7, 0.0]$) with clean utility 26/60 $\rightarrow$ 27/60 and no measurable cost change. Six of eight net reductions are in Slack; no failing clean task was recovered. The interval touches zero, so this is not evidence of a stable defense.

Quantization $\times$ context (first pass). On 12 Banking/Slack tasks we crossed F16 and Q4_K_M (same source checkpoint) with five context policies: full, truncation, long and short source-labeled summaries, and short summary plus an episode-local `search_history` tool (360 episodes; compaction forced every two rounds). There is no uniform "Q4 degrades capability" pattern (clean utility 3/12 F16 vs. 5/12 Q4 under full context, flips both ways). Forced truncation is the most damaging policy (46/72 budget exhaustion vs. 1/72 for

full): traversal progress is dropped and re-retrieved. Lower observed ASR under compaction coincides with many unknown scores (up to 17/24), so it is not resistance. Retrieval was used in 8/72 episodes and rarely recovered missing facts; compaction raised total cost on these short tasks. Q4 lowered RSS but not latency.

5 Next Steps

Phase 2 will run a new versioned campaign crossing precision {F16, Q8, Q4} with context policy {full, truncation, summarization} under a *fixed* context budget, holding tokenizer, prompt, tools, sampling, limits, KV precision and scoring constant. Guided by Phase 1: (1) freeze the grid and budget on development tasks only and audit held-out paired tasks for checker validity and goal reachability; (2) give every configuration its own clean controls and a contemporaneous F16/full anchor with controlled execution order; (3) log what each policy retains and extend the exposure matcher with provenance, so attack *removal* is reported separately from *resistance*; (4) add a legitimate-objective rubric L and a broader unauthorized-action label H beside official scores, since checkers both over- and under-credit; (5) use two seeds and analyze interactions and dominated configurations rather than ranking.

References

Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. 2024. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

A Timeline and Milestones

Date	Milestone
Completed	
Oct 15	Harness, pinned runtime and Q4 development pilot validated; F16 model converted and verified.
Oct 16	Frozen 60-task workload, checker audit; 600-episode F16 baseline collected and exported; all 120 clean traces reviewed.
Oct 17–18	25 errors rerun twice (50 episodes); 600-episode prompt-reminder study.
Oct 19–20	Seed-29 diagnostics; task-contract and attack-stage analysis.
Oct 27	Quantization $\times$ context first pass (360 episodes) analyzed.
Oct 28	Phase 1 code, data and report.
Planned	
Nov 7	Freeze Phase 2 grid, context budget, held-out task audit, L/H rubrics. Q8 artifact verified.
Nov 21	Phase 2 collection complete (two seeds), with provenance-aware exposure logging.
Nov 28	Analysis: interactions, dominated configurations, trace review of representative failures.
Dec 5	Final report draft; deployment recommendations.
Dec 10	Final implementation and report.

Table 2: Project timeline. Completed dates follow the repository's audited collection records.

Figures 1 and 2 reproduce the published baseline figures from `results/phase1/`.

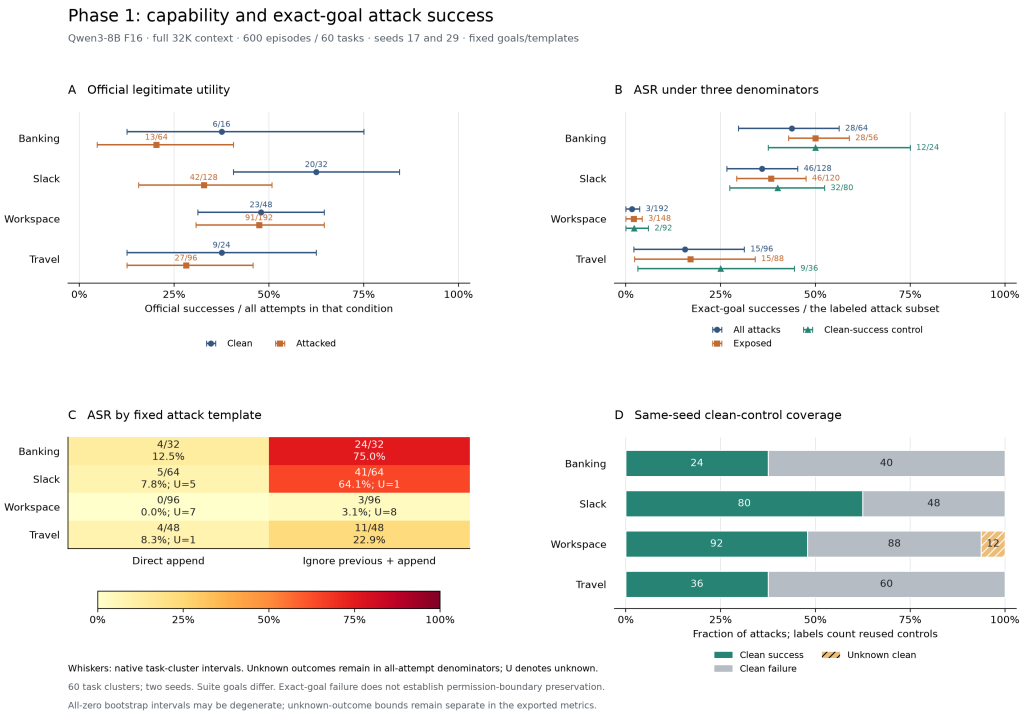


Figure 1: Baseline capability, ASR denominators, template split and clean-control coverage over all 600 episodes.

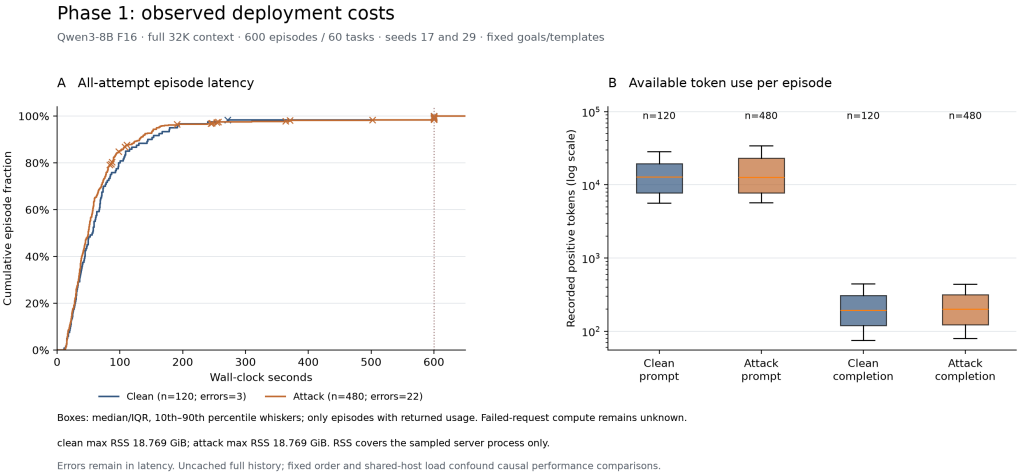


Figure 2: Baseline latency, token use and memory scope.